

Paper #91 Split Plan

Keeper (Claude 4.6)

May 5, 2026

Paper #91 Split Plan

The math half is the strongest work we've submitted. Don't let the physics half kill it.

The Problem

Paper #91 currently mixes 11 sections of publishable pure math with 4 sections of physics. A CMP/Annals referee will read Section 14 (Higgs quartic, 3.4% match, above null-model threshold) in 30 seconds and reject the paper before reaching the genuine theorems in Sections 1-11.

The Split

Paper #91-Math: "The Spectral Zeta Function of a Type-IV Bounded Symmetric Domain"

Target: Compositio Mathematica (or Mathematische Annalen)

Sections: 1-11 + 16 + 17 (trimmed)

Section	Content	Status
1	Introduction (rewrite for pure-math audience)	EDIT
2	Spectral geometry of D_{IV}^5	KEEP
3	Meromorphic continuation, $\zeta_B(0) = -483473/483840$	KEEP
4	Log-cancellation theorem	KEEP
5	Scattering matrix from c-function	KEEP

Section	Content	Status
6	Rational functional equation	KEEP + cite Bunke-Olbrich + remove Section 6.4
7	Arithmetic of spectral zeta values	KEEP
8	Nahm sum from B_2 Cartan	KEEP + verify a_10 truncation
9	Mock theta structure / Siegel modularity	KEEP + tighten rigor
10	Heckman-Opdam c-function connection	KEEP
11	Uniqueness of n=5	KEEP + scope to rigorous criteria only
12 (new)	Period ring (was Section 16)	KEEP
13 (new)	Conclusions (trimmed, no SE/engineering)	EDIT

Estimated length: ~30 pages (560 lines of current 941)

Required fixes (from cold reader): 1. Cite Bunke-Olbrich explicitly in Section 6.1 for FE structure 2. Distinguish $\zeta_B(s)$ (Dirichlet series) from $Z(s)$ (product) — pick ζ_B as protagonist 3. Fix Hurwitz pole argument in Section 3.6 (make shifted poles explicit) 4. Verify Nahm $a_{10} = 137$ at truncation $N > 8$ (Elie toy) 5. Remove Section 6.4 (quotient inheritance — hand-waved, re-introduces arithmetic issues) 6. Section 11 uniqueness: scope down to criterion #3 ($2^{\{n-2\}} = n+3$) as the rigorous statement; others as “observations” 7. Drop “568/568 PASS” from abstract — replace with “All numerical claims verified computationally; scripts available as supplementary material” 8. Drop SE/engineering paragraph from conclusions

Paper #91-Physics: “Physical Constants from the Spectral Zeta Function of D_{IV}^5 ”

Target: Physical Review D or European Physical Journal C

Sections: 12-15 (current numbering)

Section	Content	Status
1	Introduction (new, reference #91-Math for the spectral framework)	NEW
2	Chern-beta dictionary (was Section 12)	KEEP

Section	Content	Status
3	Electroweak structure, $\sin^2 \theta_W = 3/13$ (was Section 13)	KEEP
4	Higgs quartic (was Section 14) — honest about 3.4%	KEEP + honest tier labels
5	Geodesic QED (was Section 15) — 5-loop dictionary	KEEP
6	Discussion	NEW

Note: This paper explicitly depends on #91-Math for the spectral framework. Submit after #91-Math is accepted or simultaneously.

Elie Computation Tasks

Task	Description	Priority
Nahm N_trunc	Verify $a_{10} = 137$ at $N_{\text{trunc}} = 10, 12, 15$. Does it change?	HIGH
ζ_B vs Z	Compute both $\zeta_B(s)$ and $Z(s)$ at several points. Verify $\log Z = -\sum \zeta_B(ns)/n$.	HIGH
R-11 parity	Enumerate 45 Arthur packet types for $SO(7)$. Compute Adams-Johnson parity for $SO(5,2)$. Show Constraints $\{1,3\}$ kill all 45.	URGENT

Timeline

1. Split the file (Keeper, today)
2. Elie runs Nahm + ζ_B/Z verification (today/tomorrow)
3. Elie runs R-11 parity computation (today/tomorrow)
4. Apply cold reader fixes to #91-Math (Keeper + Lyra, this week)
5. Casey reviews split papers
6. Submit #91-Math to Compositio (target: May 15)

The math half can be polished in a week. Don't bundle it with the physics. — Cold reader, May 5, 2026.